

Differential calculus

Radius of curvature in Cartesian coordinates

$$\text{Radius of curvature } \rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2}$$

$$\text{Where } y_1 = \frac{dy}{dx}, y_2 = \frac{d^2y}{dx^2}$$

Find the radius of curvature of $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \dots (1)$$

Differentiating (1) with respect to x , we get

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = 0$$

$$\frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = -\frac{2}{3}x^{-\frac{1}{3}}$$

$$\frac{dy}{dx} = -\frac{x^{-\frac{1}{3}}}{y^{-\frac{1}{3}}} = -\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}$$

$$\frac{d^2y}{dx^2} = -\frac{\left(x^{\frac{1}{3}}\frac{1}{3}y^{-\frac{2}{3}}\frac{dy}{dx} - y^{\frac{1}{3}}\frac{1}{3}x^{-\frac{2}{3}}\right)}{x^{\frac{2}{3}}}$$

$$= -\frac{\left(x^{\frac{1}{3}}\frac{1}{3}y^{-\frac{2}{3}}\left(-\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}\right) - y^{\frac{1}{3}}\frac{1}{3}x^{-\frac{2}{3}}\right)}{x^{\frac{2}{3}}} = -\frac{\left(-\frac{1}{3}y^{-\frac{1}{3}} - y^{\frac{1}{3}}\frac{1}{3}x^{-\frac{2}{3}}\right)}{x^{\frac{2}{3}}}$$

$$= -\frac{\left(-\frac{1}{3y^{\frac{1}{3}}} - \frac{y^{\frac{1}{3}}}{3x^{\frac{2}{3}}}\right)}{x^{\frac{2}{3}}} = \frac{\left(x^{\frac{2}{3}} + y^{\frac{2}{3}}\right)}{3x^{\frac{2}{3}}x^{\frac{2}{3}}y^{\frac{1}{3}}}$$

$$\frac{d^2y}{dx^2} = \frac{a^{\frac{2}{3}}}{3x^{\frac{2}{3}}y^{\frac{1}{3}}} \text{ (from (1))}$$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\left(1 + \left(-\frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}\right)^2\right)^{\frac{3}{2}}}{\frac{a^{\frac{2}{3}}}{3x^{\frac{4}{3}}y^{\frac{1}{3}}}}$$

$$= 3x^{\frac{4}{3}}y^{\frac{1}{3}} \frac{\left(1 + \frac{y^{\frac{2}{3}}}{x^{\frac{2}{3}}}\right)^{\frac{3}{2}}}{a^{\frac{2}{3}}} = 3x^{\frac{4}{3}}y^{\frac{1}{3}} \frac{\left(\frac{x^{\frac{2}{3}} + y^{\frac{2}{3}}}{x^{\frac{2}{3}}}\right)^{\frac{3}{2}}}{a^{\frac{2}{3}}} = 3x^{\frac{4}{3}}y^{\frac{1}{3}} \frac{\left(a^{\frac{2}{3}}\right)^{\frac{3}{2}}}{xa^{\frac{2}{3}}} \text{ (from (1))}$$

$$= 3x^{\frac{4}{3}}y^{\frac{1}{3}} \frac{a}{xa^{\frac{2}{3}}} = 3x^{\frac{1}{3}}y^{\frac{1}{3}}a^{\frac{1}{3}}$$

$$\rho = 3(axy)^{\frac{1}{3}}$$

Find the radius of curvature at $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ on the curve $x^3 + y^3 = 3axy$

$$x^3 + y^3 = 3axy \dots (1)$$

Differentiating (1) with respect to x , we get

$$3x^2 + 3y^2 \frac{dy}{dx} = 3a \left(x \frac{dy}{dx} + y\right)$$

$$3y^2 \frac{dy}{dx} - 3ax \frac{dy}{dx} = 3ay - 3x^2 \Rightarrow (3y^2 - 3ax) \frac{dy}{dx} = 3ay - 3x^2$$

$$\frac{dy}{dx} = \frac{3ay - 3x^2}{3y^2 - 3ax}$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{3a}{2}, \frac{3a}{2}\right)} = \frac{3a\left(\frac{3a}{2}\right) - 3\left(\frac{3a}{2}\right)^2}{3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)} = -1$$

$$\frac{d^2y}{dx^2} = \frac{(3y^2 - 3ax) \left(3a \frac{dy}{dx} - 6x\right) - (3ay - 3x^2) \left(6y \frac{dy}{dx} - 3a\right)}{(3y^2 - 3ax)^2}$$

$$\left(\frac{d^2y}{dx^2}\right)_{\left(\frac{3a}{2}, \frac{3a}{2}\right)} = \frac{\left(3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)\right)\left(3a(-1) - 6\left(\frac{3a}{2}\right)\right) - \left(3a\left(\frac{3a}{2}\right) - 3\left(\frac{3a}{2}\right)^2\right)\left(6\left(\frac{3a}{2}\right)(-1) - 3a\right)}{\left(3\left(\frac{3a}{2}\right)^2 - 3a\left(\frac{3a}{2}\right)\right)^2}$$

$$= \frac{\left(\frac{27a^2}{4} - \frac{9a^2}{2}\right)(-3a - 9a) - \left(\frac{9a^2}{2} - \frac{27a^2}{4}\right)(-9a - 3a)}{\left(\frac{27a^2}{4} - \frac{9a^2}{2}\right)^2}$$

$$= \frac{\left(\frac{27a^2}{4} - \frac{9a^2}{2}\right)(-12a)(1 - (-1))}{\left(\frac{27a^2}{4} - \frac{9a^2}{2}\right)^2} = \frac{-24a}{\left(\frac{27a^2}{4} - \frac{18a^2}{4}\right)} = \frac{-24a}{\left(\frac{9a^2}{4}\right)}$$

$$\left(\frac{d^2y}{dx^2}\right)_{\left(\frac{3a}{2}, \frac{3a}{2}\right)} = \frac{-32}{3a}$$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1 + (-1)^2)^{\frac{3}{2}}}{\left(\frac{-32}{3a}\right)} = \frac{3a2^{\frac{3}{2}}}{-32} = \frac{3a2\sqrt{2}}{-32} = \frac{3a\sqrt{2}}{-16}$$

$$\rho = \frac{3a\sqrt{2}}{16} \text{ (Since radius cannot be negative)}$$

Find the radius of curvature of $x^{\frac{1}{2}} + y^{\frac{1}{2}} = a^{\frac{1}{2}}$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$

$$x^{\frac{1}{2}} + y^{\frac{1}{2}} = a^{\frac{1}{2}} \dots (1)$$

Differentiating (1) with respect to x , we get

$$\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{-\frac{1}{2}}\frac{dy}{dx} = 0$$

$$\frac{1}{2}y^{-\frac{1}{2}}\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}} = -\frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$$

$$\left(\frac{dy}{dx}\right)_{\left(\frac{1}{4}, \frac{1}{4}\right)} = -\frac{\left(\frac{1}{4}\right)^{\frac{1}{2}}}{\left(\frac{1}{4}\right)^{\frac{1}{2}}} = -1$$

$$\frac{d^2y}{dx^2} = -\frac{\left(x^{\frac{1}{2}} \frac{1}{2} y^{-\frac{1}{2}} \frac{dy}{dx} - y^{\frac{1}{2}} \frac{1}{2} x^{-\frac{1}{2}}\right)}{x}$$

$$\left(\frac{d^2y}{dx^2}\right)_{\left(\frac{1}{4}, \frac{1}{4}\right)} = -\frac{\left(\left(\frac{1}{4}\right)^{\frac{1}{2}} \frac{1}{2} \left(\frac{1}{4}\right)^{-\frac{1}{2}} (-1) - \left(\frac{1}{4}\right)^{\frac{1}{2}} \frac{1}{2} \left(\frac{1}{4}\right)^{-\frac{1}{2}}\right)}{\left(\frac{1}{4}\right)} = -\frac{\left(-\frac{1}{2} - \frac{1}{2}\right)}{\left(\frac{1}{4}\right)} = 4$$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1 + (-1)^2)^{\frac{3}{2}}}{4} = \frac{2^{\frac{3}{2}}}{4} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$$

Find the radius of curvature of $xy = c^2$ at (x, y)

$$xy = c^2 \dots (1)$$

Differentiating (1) with respect to x , we get

$$y + x \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} = -y$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\frac{d^2y}{dx^2} = -\frac{\left(x \frac{dy}{dx} - y\right)}{x^2} = -\frac{\left(x \left(-\frac{y}{x}\right) - y\right)}{x^2}$$

$$\frac{d^2y}{dx^2} = \frac{2y}{x^2}$$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\left(1 + \left(-\frac{y}{x}\right)^2\right)^{\frac{3}{2}}}{\left(\frac{2y}{x^2}\right)} = \frac{\left(\frac{x^2 + y^2}{x^2}\right)^{\frac{3}{2}}}{\left(\frac{2y}{x^2}\right)} = \frac{(x^2 + y^2)^{\frac{3}{2}}}{x^3 \left(\frac{2y}{x^2}\right)} = \frac{(x^2 + y^2)^{\frac{3}{2}}}{2xy}$$

$$\rho = \frac{(x^2 + y^2)^{\frac{3}{2}}}{2c^2} \quad \text{from (1)}$$

Find the radius of curvature of $y^2 = 4ax$ at $(at^2, 2at)$

$$y^2 = 4ax \dots (1)$$

Differentiating (1) with respect to x , we get

$$2y \frac{dy}{dx} = 4a$$

$$\frac{dy}{dx} = \frac{2a}{y}$$

$$\left(\frac{dy}{dx}\right)_{(at^2, 2at)} = \frac{2a}{2at} = \frac{1}{t}$$

$$\frac{d^2y}{dx^2} = -\frac{2a}{y^2} \frac{dy}{dx}$$

$$\left(\frac{d^2y}{dx^2}\right)_{(at^2, 2at)} = -\frac{2a}{(2at)^2} \frac{1}{t} = -\frac{1}{2at^3}$$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\left(1 + \left(\frac{1}{t}\right)^2\right)^{\frac{3}{2}}}{-\frac{1}{2at^3}} = \frac{\left(\frac{t^2 + 1}{t^2}\right)^{\frac{3}{2}}}{-\frac{1}{2at^3}} = \frac{(t^2 + 1)^{\frac{3}{2}}}{t^3 \left(-\frac{1}{2at^3}\right)}$$

$$\rho = 2a(t^2 + 1)^{\frac{3}{2}} \quad (\text{Since the radius of curvature is non-negative})$$

Find the radius of curvature of $\left(\frac{x}{a}\right)^{\frac{1}{2}} + \left(\frac{y}{b}\right)^{\frac{1}{2}} = 1$ at (x, y)

$$\left(\frac{x}{a}\right)^{\frac{1}{2}} + \left(\frac{y}{b}\right)^{\frac{1}{2}} = 1 \dots (1)$$

Differentiating (1) with respect to x , we get

$$\frac{1}{2\sqrt{a}} x^{-\frac{1}{2}} + \frac{1}{2\sqrt{b}} y^{-\frac{1}{2}} \frac{dy}{dx} = 0$$

$$\frac{1}{2\sqrt{b}} y^{-\frac{1}{2}} \frac{dy}{dx} = -\frac{1}{2\sqrt{a}} x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{2\sqrt{b}x^{-\frac{1}{2}}}{2\sqrt{a}y^{-\frac{1}{2}}} = -\frac{\sqrt{b}y^{\frac{1}{2}}}{\sqrt{a}x^{\frac{1}{2}}}$$

$$\frac{d^2y}{dx^2} = -\frac{\sqrt{b}}{\sqrt{a}} \left[\frac{\left(x^{\frac{1}{2}} \frac{1}{2} y^{-\frac{1}{2}} \frac{dy}{dx} - y^{\frac{1}{2}} \frac{1}{2} x^{-\frac{1}{2}} \right)}{x} \right]$$

$$\frac{d^2y}{dx^2} = -\frac{\sqrt{b}}{\sqrt{a}} \left[\frac{\left(x^{\frac{1}{2}} \frac{1}{2} y^{-\frac{1}{2}} \left(-\frac{\sqrt{b}y^{\frac{1}{2}}}{\sqrt{a}x^{\frac{1}{2}}} \right) - y^{\frac{1}{2}} \frac{1}{2} x^{-\frac{1}{2}} \right)}{x} \right] = -\frac{\sqrt{b}}{\sqrt{a}} \left[\frac{\left(\frac{1}{2} \left(-\frac{\sqrt{b}}{\sqrt{a}} \right) - \frac{1}{2} \frac{\sqrt{y}}{\sqrt{x}} \right)}{x} \right]$$

$$= \frac{\sqrt{b}}{\sqrt{a}} \left[\frac{(\sqrt{bx} + \sqrt{ay})}{2\sqrt{axx}} \right]$$

$$\begin{aligned} \rho &= \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{\left(1 + \left(-\frac{\sqrt{b}y^{\frac{1}{2}}}{\sqrt{a}x^{\frac{1}{2}}} \right)^2 \right)^{\frac{3}{2}}}{\frac{\sqrt{b}}{\sqrt{a}} \left[\frac{(\sqrt{bx} + \sqrt{ay})}{2\sqrt{axx}} \right]} = \frac{\left(1 + \left(\frac{by}{ax} \right) \right)^{\frac{3}{2}}}{\frac{\sqrt{b}}{\sqrt{a}} \left[\frac{(\sqrt{bx} + \sqrt{ay})}{2\sqrt{axx}} \right]} \\ &= \frac{(ax + by)^{\frac{3}{2}}}{(ax)^{\frac{3}{2}} \frac{\sqrt{b}}{\sqrt{a}} \left[\frac{(\sqrt{bx} + \sqrt{ay})}{2\sqrt{axx}} \right]} \end{aligned}$$

$$\rho = \frac{(ax + by)^{\frac{3}{2}}}{(ax)^{\frac{3}{2}} \frac{\sqrt{b}}{\sqrt{a}} \left[\frac{(\sqrt{b}\sqrt{a})}{2\sqrt{axx}} \right]} \quad (\text{from (1)})$$

$$\rho = \frac{2(ax + by)^{\frac{3}{2}}}{ab}$$

Radius of curvature in Parametric form

IF $x = f(t), y = g(t)$

$$\text{Radius of curvature } \rho = \frac{((f')^2 + (g')^2)^{\frac{3}{2}}}{f'g'' - g'f''}$$

Where $f' = \frac{dx}{dt}, f'' = \frac{d^2x}{dt^2}, g' = \frac{dy}{dt}, g'' = \frac{d^2y}{dt^2}$

Find the radius of curvature of $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

The parametric form is $x = a \cos^3 \theta, y = a \sin^3 \theta \dots (1)$

$$f' = \frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta) = -3a \cos^2 \theta \sin \theta$$

$$f'' = \frac{d^2x}{d\theta^2} = -3a \cos^2 \theta \cos \theta - 6a \cos \theta (-\sin \theta) \sin \theta = -3a \cos^3 \theta + 6a \cos \theta \sin^2 \theta$$

$$g' = \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$g'' = \frac{d^2y}{d\theta^2} = 3a \sin^2 \theta (-\sin \theta) + 6a \sin \theta \cos \theta \cos \theta = -3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta$$

$$\text{Radius of curvature } \rho = \frac{((f')^2 + (g')^2)^{\frac{3}{2}}}{f'g'' - g'f''}$$

$$= \frac{((-3a \cos^2 \theta \sin \theta)^2 + (3a \sin^2 \theta \cos \theta)^2)^{\frac{3}{2}}}{(-3a \cos^2 \theta \sin \theta)(-3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta) - (3a \sin^2 \theta \cos \theta)(-3a \cos^3 \theta + 6a \cos \theta \sin^2 \theta)}$$

$$= \frac{(3^2 a^2 \cos^4 \theta \sin^2 \theta + 3^2 a^2 \cos^2 \theta \sin^4 \theta)^{\frac{3}{2}}}{(9a^2 \cos^2 \theta \sin^2 \theta)(\sin^2 \theta - 2 \cos^2 \theta + \cos^2 \theta - 2 \sin^2 \theta)}$$

$$= \frac{(3^2 a^2 \cos^2 \theta \sin^2 \theta)^{\frac{3}{2}} (\cos^2 \theta + \sin^2 \theta)^{\frac{3}{2}}}{(9a^2 \cos^2 \theta \sin^2 \theta)(-\sin^2 \theta - \cos^2 \theta)}$$

$$= \frac{(3^3 a^3 \cos^3 \theta \sin^3 \theta)(\cos^2 \theta + \sin^2 \theta)^{\frac{3}{2}}}{-(9a^2 \cos^2 \theta \sin^2 \theta)(\sin^2 \theta + \cos^2 \theta)}$$

$$\rho = -3a \cos \theta \sin \theta$$

$$\rho = 3a \cos \theta \sin \theta \quad (\text{Since radius cannot be negative})$$

$$\rho = 3a \left(\frac{x}{a}\right)^{\frac{1}{3}} \left(\frac{y}{a}\right)^{\frac{1}{3}} = 3(xy)^{\frac{1}{3}}$$

Find the radius of curvature of $y^2 = 4ax$

The parametric form is $x = at^2, y = 2at \dots (1)$

$$f' = \frac{dx}{dt} = 2at, f'' = \frac{d^2x}{dt^2} = 2a, g' = \frac{dy}{dt} = 2a, g'' = \frac{d^2y}{dt^2} = 0$$

$$\text{Radius of curvature } \rho = \frac{((f')^2 + (g')^2)^{\frac{3}{2}}}{f'g'' - g'f''}$$

$$= \frac{((2at)^2 + (2a)^2)^{\frac{3}{2}}}{(2at)(0) - (2a)(2a)} = \frac{(2a)^3(t^2 + 1)^{\frac{3}{2}}}{-(2a)^2}$$

$$\rho = 2a(t^2 + 1)^{\frac{3}{2}} \text{ (Since radius cannot be negative)}$$

Find the radius of curvature of $x^2 = 4ay$

The parametric form is $x = 2at, y = at^2 \dots (1)$

$$f' = \frac{dx}{dt} = 2a, f'' = \frac{d^2x}{dt^2} = 0, g' = \frac{dy}{dt} = 2at, g'' = \frac{d^2y}{dt^2} = 2a$$

$$\text{Radius of curvature } \rho = \frac{((f')^2 + (g')^2)^{\frac{3}{2}}}{f'g'' - g'f''}$$

$$= \frac{((2a)^2 + (2at)^2)^{\frac{3}{2}}}{(2a)(2a) - (2at)(0)} = \frac{(2a)^3(1 + t^2)^{\frac{3}{2}}}{(2a)^2}$$

$$\rho = 2a(1 + t^2)^{\frac{3}{2}}$$

Centre of Curvature in Cartesian form:

$$\bar{X} = x - \frac{y_1(1 + (y_1)^2)}{y_2}$$

$$\bar{Y} = y + \frac{(1 + (y_1)^2)}{y_2}$$

Circle of curvature

$$(x - \bar{X})^2 + (y - \bar{Y})^2 = \rho^2$$

Find the circle of the curve $xy = c^2$ at (c, c)

$$xy = c^2 \dots (1)$$

Differentiating (1) with respect to x , we get

$$xy_1 + y = 0 \Rightarrow y_1 = -\frac{y}{x}$$

$$(y_1)_{(c,c)} = -\frac{c}{c} = -1$$

$$y_2 = -\frac{xy_1 - y}{x^2}$$

$$(y_2)_{(c,c)} = -\frac{c(-1) - c}{c^2} = \frac{2c}{c^2} = \frac{2}{c}$$

$$\bar{X} = x - \frac{y_1(1 + (y_1)^2)}{y_2}$$

$$\bar{X} = c - \frac{-1(1 + (-1)^2)}{\left(\frac{2}{c}\right)} = c + \frac{2c}{2} = 2c$$

$$\bar{Y} = y + \frac{(1 + (y_1)^2)}{y_2}$$

$$\bar{Y} = c + \frac{(1 + (-1)^2)}{\left(\frac{2}{c}\right)} = c + \frac{2c}{2} = 2c$$

The center of curvature is $(2c, 2c)$

$$\rho = \frac{(1 + y_1^2)^{\frac{3}{2}}}{y_2} = \frac{(1 + (-1)^2)^{\frac{3}{2}}}{\left(\frac{2}{c}\right)} = \frac{2^{\frac{3}{2}}c}{2} = \sqrt{2}c$$

The circle of curvature is given by

$$(x - \bar{X})^2 + (y - \bar{Y})^2 = \rho^2$$

$$(x - 2c)^2 + (y - 2c)^2 = 2c^2$$

Centre of Curvature in parametric form:

IF $x = f(t), y = g(t)$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

where $\dot{x} = \frac{dx}{dt}, \ddot{x} = \frac{d^2x}{dt^2}, \dot{y} = \frac{dy}{dt}, \ddot{y} = \frac{d^2y}{dt^2}$

Find the equation of the evolute of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Solution:

The parametric form is $x = a \cos^3 \theta, y = a \sin^3 \theta \dots (1)$

$$\dot{x} = \frac{dx}{d\theta} = 3a \cos^2 \theta (-\sin \theta) = -3a \cos^2 \theta \sin \theta$$

$$\ddot{x} = \frac{d^2x}{d\theta^2} = -3a \cos^2 \theta \cos \theta - 6a \cos \theta (-\sin \theta) \sin \theta = -3a \cos^3 \theta + 6a \cos \theta \sin^2 \theta$$

$$\dot{y} = \frac{dy}{d\theta} = 3a \sin^2 \theta \cos \theta$$

$$\ddot{y} = \frac{d^2y}{d\theta^2} = 3a \sin^2 \theta (-\sin \theta) + 6a \sin \theta \cos \theta \cos \theta = -3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= a \cos^3 \theta - \frac{3a \sin^2 \theta \cos \theta ((-3a \cos^2 \theta \sin \theta)^2 + (3a \sin^2 \theta \cos \theta)^2)}{(-3a \cos^2 \theta \sin \theta)(-3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta) - (3a \sin^2 \theta \cos \theta)(-3a \cos^3 \theta + 6a \cos \theta \sin^2 \theta)}$$

$$= a \cos^3 \theta - \frac{3a \sin^2 \theta \cos \theta (3^2 a^2 \cos^4 \theta \sin^2 \theta + 3^2 a^2 \cos^2 \theta \sin^4 \theta)}{(9a^2 \cos^2 \theta \sin^2 \theta)(\sin^2 \theta - 2 \cos^2 \theta + \cos^2 \theta - 2 \sin^2 \theta)}$$

$$= a \cos^3 \theta - \frac{(3a \sin^2 \theta \cos \theta) 3^2 a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)}{(9a^2 \cos^2 \theta \sin^2 \theta)(-\sin^2 \theta - \cos^2 \theta)}$$

$$= a \cos^3 \theta - \frac{(3a \sin^2 \theta \cos \theta)}{-(\sin^2 \theta + \cos^2 \theta)}$$

$$\bar{X} = a \cos^3 \theta + 3a \sin^2 \theta \cos \theta$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= a \sin^3 \theta + \frac{-3a \cos^2 \theta \sin \theta ((-3a \cos^2 \theta \sin \theta)^2 + (3a \sin^2 \theta \cos \theta)^2)}{(-3a \cos^2 \theta \sin \theta)(-3a \sin^3 \theta + 6a \sin \theta \cos^2 \theta) - (3a \sin^2 \theta \cos \theta)(-3a \cos^3 \theta + 6a \cos \theta \sin^2 \theta)}$$

$$= a \sin^3 \theta + \frac{-3a \cos^2 \theta \sin \theta (3^2 a^2 \cos^4 \theta \sin^2 \theta + 3^2 a^2 \cos^2 \theta \sin^4 \theta)}{(9a^2 \cos^2 \theta \sin^2 \theta)(\sin^2 \theta - 2 \cos^2 \theta + \cos^2 \theta - 2 \sin^2 \theta)}$$

$$= a \sin^3 \theta + \frac{(-3 a \cos^2 \theta \sin \theta) 3^2 a^2 \cos^2 \theta \sin^2 \theta (\cos^2 \theta + \sin^2 \theta)}{(9 a^2 \cos^2 \theta \sin^2 \theta)(-\sin^2 \theta - \cos^2 \theta)}$$

$$= a \sin^3 \theta + \frac{(-3 a \cos^2 \theta \sin \theta)}{-(\sin^2 \theta + \cos^2 \theta)}$$

$$\bar{Y} = a \sin^3 \theta + 3 a \cos^2 \theta \sin \theta$$

$$\bar{X} + \bar{Y} = a \cos^3 \theta + 3 a \sin^2 \theta \cos \theta + a \sin^3 \theta + 3 a \cos^2 \theta \sin \theta$$

$$\bar{X} + \bar{Y} = a(\cos \theta + \sin \theta)^3 \Rightarrow (\bar{X} + \bar{Y})^{\frac{1}{3}} = a^{\frac{1}{3}}(\cos \theta + \sin \theta) \dots (2)$$

$$\bar{X} - \bar{Y} = a \cos^3 \theta + 3 a \sin^2 \theta \cos \theta - a \sin^3 \theta - 3 a \cos^2 \theta \sin \theta$$

$$\bar{X} - \bar{Y} = a(\cos \theta - \sin \theta)^3 \Rightarrow (\bar{X} - \bar{Y})^{\frac{1}{3}} = a^{\frac{1}{3}}(\cos \theta - \sin \theta) \dots (3)$$

(2)² + (3)² gives

$$\begin{aligned} (\bar{X} + \bar{Y})^{\frac{2}{3}} + (\bar{X} - \bar{Y})^{\frac{2}{3}} &= a^{\frac{2}{3}}(\cos \theta + \sin \theta)^2 + a^{\frac{2}{3}}(\cos \theta - \sin \theta)^2 \\ &= a^{\frac{2}{3}}(\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta - 2 \sin \theta \cos \theta) \end{aligned}$$

$$(\bar{X} + \bar{Y})^{\frac{2}{3}} + (\bar{X} - \bar{Y})^{\frac{2}{3}} = 2a^{\frac{2}{3}} \dots (4)$$

The locus of (4) is the evolute of the given curve

$$(x + y)^{\frac{2}{3}} + (x - y)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$$

Find the equation of the evolute of the curve $y^2 = 4ax$.

Solution:

The parametric form is $x = at^2, y = 2at \dots (1)$

$$\dot{x} = \frac{dx}{dt} = 2at, \ddot{x} = \frac{d^2x}{dt^2} = 2a, \dot{y} = \frac{dy}{dt} = 2a, \ddot{y} = \frac{d^2y}{dt^2} = 0$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{X} = at^2 - \frac{2a((2at)^2 + (2a)^2)}{(2at)(0) - (2a)(2a)} = at^2 + \frac{((2a)^3(t^2 + 1))}{(2a)^2}$$

$$\bar{X} = at^2 + 2a(t^2 + 1) = at^2 + 2at^2 + 2a = 3at^2 + 2a$$

$$\bar{X} - 2a = 3at^2 \Rightarrow \frac{\bar{X} - 2a}{3a} = t^2 \Rightarrow t = \left(\frac{\bar{X} - 2a}{3a} \right)^{\frac{1}{2}} \dots (2)$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{Y} = 2at + \frac{2at((2at)^2 + (2a)^2)}{(2at)(0) - (2a)(2a)} = 2at - \frac{((2a)^3 t(t^2 + 1))}{(2a)^2}$$

$$\bar{Y} = 2at - 2at(t^2 + 1) = 2at - 2at^3 - 2at = -2at^3$$

$$\bar{Y} = -2at^3 \Rightarrow \frac{\bar{Y}}{-2a} = t^3 \Rightarrow t = \left(\frac{\bar{Y}}{-2a} \right)^{\frac{1}{3}} \dots (3)$$

The center of curvature is $(3at^2 + 2a, -2at^3)$

From (2) and (3), we get

$$\left(\frac{\bar{Y}}{-2a} \right)^{\frac{1}{3}} = \left(\frac{\bar{X} - 2a}{3a} \right)^{\frac{1}{2}}$$

Raising the powers with 6 on both sides, we get

$$\left(\frac{\bar{Y}}{-2a} \right)^2 = \left(\frac{\bar{X} - 2a}{3a} \right)^3 \Rightarrow \frac{\bar{Y}^2}{4a^2} = \frac{(\bar{X} - 2a)^3}{27a^3}$$

$$27a\bar{Y}^2 = 4(\bar{X} - 2a)^3$$

The locus of (4) is the evolute of the given curve

$$27ay^2 = 4(x - 2a)^3$$

Find the equation of the evolute of the curve $x^2 = 4ay$.

Solution:

The parametric form is $x = 2at, y = at^2 \dots (1)$

$$\dot{x} = \frac{dx}{dt} = 2a, \ddot{x} = \frac{d^2x}{dt^2} = 0, \dot{y} = \frac{dy}{dt} = 2at, \ddot{y} = \frac{d^2y}{dt^2} = 2a$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{X} = 2at - \frac{2at((2a)^2 + (2at)^2)}{(2a)(2a) - (2at)(0)} = 2at - \frac{((2a)^3 t(1 + t^2))}{(2a)^2}$$

$$\bar{X} = 2at - 2at(t^2 + 1) = 2at - 2at^3 - 2at = -2at^3$$

$$\bar{X} = -2at^3 \Rightarrow \frac{\bar{X}}{-2a} = t^3 \Rightarrow t = \left(\frac{\bar{X}}{-2a} \right)^{\frac{1}{3}} \dots (2)$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{Y} = at^2 + \frac{2a((2at)^2 + (2a)^2)}{(2a)(2a) - (2at)(0)} = 2at + \frac{((2a)^3(t^2 + 1))}{(2a)^2}$$

$$\bar{Y} = at^2 + 2a(t^2 + 1) = at^2 + 2at^2 + 2a = 3at^2 + 2a$$

$$\bar{Y} - 2a = 3at^2 \Rightarrow \frac{\bar{Y} - 2a}{3a} = t^2 \Rightarrow t = \left(\frac{\bar{Y} - 2a}{3a} \right)^{\frac{1}{2}} \dots (3)$$

The center of curvature is $(-2at^3, 3at^2 + 2a)$

From (2) and (3), we get

$$\left(\frac{\bar{X}}{-2a} \right)^{\frac{1}{3}} = \left(\frac{\bar{Y} - 2a}{3a} \right)^{\frac{1}{2}}$$

Raising the powers with 6 on both sides, we get

$$\left(\frac{\bar{X}}{-2a} \right)^2 = \left(\frac{\bar{Y} - 2a}{3a} \right)^3 \Rightarrow \frac{\bar{X}^2}{4a^2} = \frac{(\bar{Y} - 2a)^3}{27a^3}$$

$$27a\bar{X}^2 = 4(\bar{Y} - 2a)^3$$

The locus of (4) is the evolute of the given curve

$$27ax^2 = 4(y - 2a)^3$$

Find the equation of the evolute of the curve $xy = c^2$.

Solution:

The parametric form is $x = ct, y = \frac{c}{t} \dots (1)$

$$\dot{x} = \frac{dx}{dt} = c, \ddot{x} = \frac{d^2x}{dt^2} = 0, \dot{y} = \frac{dy}{dt} = -\frac{c}{t^2}, \ddot{y} = \frac{d^2y}{dt^2} = \frac{2c}{t^3}$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{X} = ct - \frac{\left(-\frac{c}{t^2}\right)\left((c)^2 + \left(-\frac{c}{t^2}\right)^2\right)}{(c)\left(\frac{2c}{t^3}\right) - \left(-\frac{c}{t^2}\right)(0)} = ct + \frac{\left(\frac{c^3}{t^2}\right)\left(1 + \frac{1}{t^4}\right)}{(c)\left(\frac{2c}{t^3}\right)}$$

$$\bar{X} = ct + \frac{ct\left(1 + \frac{1}{t^4}\right)}{2} = \frac{2ct + ct + \frac{c}{t^3}}{2} = \frac{3ct + \frac{c}{t^3}}{2}$$

$$\bar{X} = \frac{c}{2}\left(3t + \frac{1}{t^3}\right)$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$\bar{Y} = \frac{c}{t} + \frac{(c)\left((c)^2 + \left(-\frac{c}{t^2}\right)^2\right)}{(c)\left(\frac{2c}{t^3}\right) - \left(-\frac{c}{t^2}\right)(0)} = \frac{c}{t} + \frac{ct^3\left(1 + \frac{1}{t^4}\right)}{2}$$

$$\bar{Y} = \frac{2c + ct^4 + c}{2t} = \frac{ct^4 + 3c}{2t} = \frac{c}{2}\left(t^3 + \frac{3}{t}\right)$$

The center of curvature is $\left(\frac{c}{2}\left(3t + \frac{1}{t^3}\right), \frac{c}{2}\left(t^3 + \frac{3}{t}\right)\right)$

$$\bar{X} + \bar{Y} = \frac{c}{2}\left(3t + \frac{1}{t^3} + t^3 + \frac{3}{t}\right) = \frac{c}{2}\left(t^3 + 3t + \frac{3}{t} + \frac{1}{t^3}\right) = \frac{c}{2}\left(t + \frac{1}{t}\right)^3$$

$$(\bar{X} + \bar{Y})^{\frac{1}{3}} = \left(\frac{c}{2}\right)^{\frac{1}{3}}\left(t + \frac{1}{t}\right) \dots (2)$$

$$\bar{X} - \bar{Y} = \frac{c}{2}\left(3t + \frac{1}{t^3} - t^3 - \frac{3}{t}\right) = -\frac{c}{2}\left(t^3 - 3t + \frac{3}{t} - \frac{1}{t^3}\right) = -\frac{c}{2}\left(t - \frac{1}{t}\right)^3$$

$$(\bar{X} - \bar{Y})^{\frac{1}{3}} = \left(-\frac{c}{2}\right)^{\frac{1}{3}}\left(t - \frac{1}{t}\right) \dots (3)$$

$(2)^2 - (3)^2$ gives

$$(\bar{X} + \bar{Y})^{\frac{2}{3}} - (\bar{X} - \bar{Y})^{\frac{2}{3}} = \left(\frac{c}{2}\right)^{\frac{2}{3}} \left(t + \frac{1}{t}\right)^2 - \left(-\frac{c}{2}\right)^{\frac{2}{3}} \left(t - \frac{1}{t}\right)^2$$

$$(\bar{X} + \bar{Y})^{\frac{2}{3}} - (\bar{X} - \bar{Y})^{\frac{2}{3}} = \left(\frac{c}{2}\right)^{\frac{2}{3}} \left[\left(t + \frac{1}{t}\right)^2 - \left(t - \frac{1}{t}\right)^2 \right] = \left(\frac{c}{2}\right)^{\frac{2}{3}} \left(t^2 + 2 + \frac{1}{t^2} - \left(t^2 - 2 + \frac{1}{t^2}\right)\right)$$

$$(\bar{X} + \bar{Y})^{\frac{2}{3}} - (\bar{X} - \bar{Y})^{\frac{2}{3}} = 4 \left(\frac{c}{2}\right)^{\frac{2}{3}}$$

$$(\bar{X} + \bar{Y})^{\frac{2}{3}} - (\bar{X} - \bar{Y})^{\frac{2}{3}} = (4c)^{\frac{2}{3}}$$

The locus of (4) is the evolute of the given curve

$$(x + y)^{\frac{2}{3}} - (x - y)^{\frac{2}{3}} = (4c)^{\frac{2}{3}}$$

Find the equation of the evolute of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

The parametric form is $x = a \cos \theta, y = b \sin \theta \dots (1)$

$$\dot{x} = \frac{dx}{d\theta} = -a \sin \theta, \ddot{x} = \frac{d^2x}{d\theta^2} = -a \cos \theta, \dot{y} = \frac{dy}{d\theta} = b \cos \theta, \ddot{y} = \frac{d^2y}{d\theta^2} = -b \sin \theta$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= a \cos \theta - \frac{b \cos \theta ((-a \sin \theta)^2 + (b \cos \theta)^2)}{(-a \sin \theta)(-b \sin \theta) - (b \cos \theta)(-a \cos \theta)}$$

$$= a \cos \theta - \frac{b \cos \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{ab \sin^2 \theta + ab \cos^2 \theta}$$

$$= a \cos \theta - \frac{b \cos \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{ab (\sin^2 \theta + \cos^2 \theta)}$$

$$= a \cos \theta - \frac{\cos \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{a} = \frac{a^2 \cos \theta - \cos \theta (a^2 (1 - \cos^2 \theta) + b^2 \cos^2 \theta)}{a}$$

$$\bar{X} = \frac{a^2 \cos \theta - a^2 \cos \theta + (a^2 \cos^3 \theta - b^2 \cos^3 \theta)}{a} = \frac{(a^2 - b^2) \cos^3 \theta}{a}$$

$$a\bar{X} = (a^2 - b^2) \cos^3 \theta \Rightarrow \frac{a\bar{X}}{a^2 - b^2} = \cos^3 \theta \Rightarrow \left(\frac{a\bar{X}}{a^2 - b^2} \right)^{\frac{1}{3}} = \cos \theta \dots (2)$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= b \sin \theta + \frac{-a \sin \theta ((-a \sin \theta)^2 + (b \cos \theta)^2)}{(-a \sin \theta)(-b \sin \theta) - (b \cos \theta)(-a \cos \theta)}$$

$$= b \sin \theta - \frac{a \sin \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{ab \sin^2 \theta + ab \cos^2 \theta}$$

$$= b \sin \theta - \frac{a \sin \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{ab (\sin^2 \theta + \cos^2 \theta)}$$

$$= b \sin \theta - \frac{\sin \theta (a^2 \sin^2 \theta + b^2 \cos^2 \theta)}{b} = \frac{b^2 \sin \theta - \sin \theta (a^2 \sin^2 \theta + b^2 (1 - \sin^2 \theta))}{b}$$

$$\bar{Y} = \frac{b^2 \sin \theta - a^2 \sin^3 \theta + b^2 \sin \theta + b^2 \sin^3 \theta}{b} = \frac{-(a^2 - b^2) \sin^3 \theta}{b}$$

$$b\bar{Y} = -(a^2 - b^2) \sin^3 \theta \Rightarrow \frac{-b\bar{Y}}{a^2 - b^2} = \sin^3 \theta \Rightarrow \left(\frac{-b\bar{Y}}{a^2 - b^2} \right)^{\frac{1}{3}} = \sin \theta \dots (3)$$

(2)² + (3)² gives

$$\left(\frac{a\bar{X}}{a^2 - b^2} \right)^{\frac{2}{3}} + \left(\frac{-b\bar{Y}}{a^2 - b^2} \right)^{\frac{2}{3}} = \cos^2 \theta + \sin^2 \theta = 1$$

$$(a\bar{X})^{\frac{2}{3}} + (b\bar{Y})^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}} \dots (4)$$

The locus of (4) is the evolute of the given curve

$$(ax)^{\frac{2}{3}} + (by)^{\frac{2}{3}} = (a^2 - b^2)^{\frac{2}{3}}$$

Find the equation of the evolute of the curve $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$

Solution:

$$x = a(\theta - \sin \theta), y = a(1 - \cos \theta) \dots (1)$$

$$\dot{x} = \frac{dx}{d\theta} = a(1 - \cos \theta), \ddot{x} = \frac{d^2x}{d\theta^2} = a \sin \theta, \dot{y} = \frac{dy}{d\theta} = a \sin \theta, \ddot{y} = \frac{d^2y}{d\theta^2} = a \cos \theta$$

$$\bar{X} = x - \frac{\dot{y}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= a(\theta - \sin \theta) - \frac{a \sin \theta ((a(1 - \cos \theta))^2 + (a \sin \theta)^2)}{a(1 - \cos \theta)(a \cos \theta) - (a \sin \theta)(a \sin \theta)}$$

$$= a(\theta - \sin \theta) - \frac{a \sin \theta (a^2 + a^2 \cos^2 \theta - 2a^2 \cos \theta + a^2 \sin^2 \theta)}{a^2 \cos \theta - a^2 \cos^2 \theta - a^2 \sin^2 \theta}$$

$$= a\theta - a \sin \theta - \frac{a \sin \theta (a^2 + a^2 - 2a^2 \cos \theta)}{a^2 \cos \theta - a^2} = a\theta - a \sin \theta - \frac{a \sin \theta (2a^2 - 2a^2 \cos \theta)}{a^2 \cos \theta - a^2}$$

$$= a\theta - a \sin \theta + \frac{2a \sin \theta (a^2 \cos \theta - a^2)}{a^2 \cos \theta - a^2} = a\theta - a \sin \theta + 2a \sin \theta$$

$$\bar{X} = a\theta + a \sin \theta \dots (2)$$

$$\bar{Y} = y + \frac{\dot{x}((\dot{x})^2 + (\dot{y})^2)}{\dot{x}\ddot{y} - \dot{y}\ddot{x}}$$

$$= a(1 - \cos \theta) + \frac{a(1 - \cos \theta) ((a(1 - \cos \theta))^2 + (a \sin \theta)^2)}{a(1 - \cos \theta)(a \cos \theta) - (a \sin \theta)(a \sin \theta)}$$

$$= a(1 - \cos \theta) + \frac{a(1 - \cos \theta)(a^2 + a^2 \cos^2 \theta - 2a^2 \cos \theta + a^2 \sin^2 \theta)}{a^2 \cos \theta - a^2 \cos^2 \theta - a^2 \sin^2 \theta}$$

$$= a(1 - \cos \theta) + \frac{a(1 - \cos \theta)(a^2 + a^2 - 2a^2 \cos \theta)}{a^2 \cos \theta - a^2}$$

$$= a(1 - \cos \theta) + \frac{a(1 - \cos \theta)(2a^2 - 2a^2 \cos \theta)}{a^2 \cos \theta - a^2}$$

$$= a(1 - \cos \theta) - \frac{2a(1 - \cos \theta)(a^2 \cos \theta - a^2)}{a^2 \cos \theta - a^2} = a(1 - \cos \theta) - 2a(1 - \cos \theta)$$

$$\bar{Y} = -a(1 - \cos \theta) \dots (3)$$

The locus of (2) and (3) is the evolute of the given curve

$$x = a\theta + a \sin \theta, y = -a(1 - \cos \theta)$$